

PAOLA CALLE

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EDUCATION

New York University, New York, NY
Master of Science in Data Science, **GPA**: 3.45

Expected May 2027

Smith College, Northampton, MA

June 2025

Bachelor of Science in Computer Science and Statistical Data Science, **GPA**: 3.99

Relevant Coursework: Machine Learning, Databases, Big Data, Statistical Testing

PROFESSIONAL EXPERIENCE

MTA, New York, NY

January 2026 - Present

Data Analyst (OCC), Emerging Talent Intern

- Developed interactive, KPI-driven Power BI reports, Power Apps dashboards, and GIS-based visualizations to support operational and decision-making workflows.
- Designed an internal GenAI-enabled search and insights application that allows users to ask natural-language questions and quickly retrieve relevant operational records and supporting documents.
- Implemented a RAG pipeline with an indexing layer, vector retrieval, and API integrations to surface operational data across multiple systems.
- Integrated user feedback loops and interaction logging to iteratively refine retrieval accuracy, ranking, and response quality.

Elliot Investment Management L.P., New York, NY

June 2024 – August 2024

Application Developer Intern (Hybrid)

- Engineered scalable backend services in C#/.NET to support dynamic schema evolution for enterprise-wide data systems.
- Designed RESTful APIs enabling real-time data access and integration.
- Built CI/CD pipelines using GitHub Actions, Octopus, and Kubernetes to automate testing.
- Developed read-optimized data views to support historical feature tracking and model input consistency.

Computer Science Department, Smith College, Northampton, MA

June 2023 – September 2025

Machine Learning & Networks Research Assistant

- Designed and optimized anomaly detection models (OCSVM, XGBoost, Isolation Forest) for large-scale DNS traffic classification.
- Architected a hybrid modeling framework combining heuristic structural signals (HTML body size, tag distribution, formatting patterns) with transformer-based NLP models to capture both semantic and structural anomalies.
- Fine-tuned pre-trained BERT models in TensorFlow for contextual pattern detection in structured and semi-structured HTML data.
- Developed an ensemble aggregation strategy (majority voting + averaged confidence scoring) to improve robustness and reduce false positives across heterogeneous models.
- Built scalable end-to-end ELT pipeline (AWS S3 ingestion, JSON parsing, feature engineering, multi-threaded preprocessing, batch training) to support rapid experimentation and model iteration.

PROJECTS

Wavvy Music AI, New York, NY

September 2025 – December 2025

Data Engineer / Full-Stack Developer — Independent Project

- Built an ETL pipeline to ingest Spotify Web API data into a normalized PostgreSQL schema for analytics and machine learning feature generation.
- Designed scalable data models to support recommendation experimentation and predictive modeling workflows.
- Deployed a React-based frontend and Flask API backend, with automated CI/CD pipelines using GitHub Actions.
- Developed music recommendation and semantic mood analysis models to surface personalized track suggestions based on audio features and contextual similarity.

SKILLS

Programming: Python, R, C++, C#, SQL, JavaScript; **Machine Learning**: Scikit-learn, TensorFlow, PyTorch, XGBoost, Feature Engineering, Model Evaluation; **Generative AI**: LLM APIs, Prompt Engineering, Model Benchmarking; **Data Engineering**: ETL/ELT Pipelines, AWS, JSON Processing, Feature Stores; **MLOps & DevOps**: CI/CD (GitHub Actions), Kubernetes, REST APIs, Automated Testing, OpenTelemetry; **Visualization**: Power BI, Tableau; **Databases**: PostgreSQL, MySQL.